CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Civil Examination

May 2011

CL201 SURVEYING

Date: 17.5.2011, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Enlist the methods of plane table surveying. Describe the method of orienting plane table by backsighting. **04**

(b) What is the application of total station in modern surveying? **03**

Q - 2 (a) Explain the process of setting out of small. **04**

(b) What is meant by degree of curve? Derive its relationship with the radius of curve. **04**

(c) Two straight lines intersect at a chainage of 1250.50m, the angle of intersection is 80°. If the degree of the curve is 5°, by using 30m chain, determine the following **06**

a) Radius of curve
b) Tangent distance
c) Length of curve
d) Length of long chord
e) Chainage of point of curvature
f) Apex distance and the mid-ordinate

OR

Q - 2 (a) Draw a neat sketch of simple circular curve and show following elements. **04**

a) Long chord
b) Point of commencement
c) Mid-ordinate
d) Rear tangent

(b) What is the use of slope rail and boning rod in setting out works. **04**

(c) Two straights AB & BC intersect at a chainage of 3450m. The angle of intersection is 130°. It is required to set out 6° simple circular curve to connect the straights. Calculate all the data necessary to set out the curve by method of **offset from chord produced** with an interval of 30m. **06**

Q - 3 (a) **Define the following** **02**

a) Apex distance
b) Fiducial edge of the alidade
c) Compound curve
d) U- fork

(b) What are the advantages and disadvantages of plane table survey? **05**

(c) Discuss in details the steps involved in setting out centre line of tunnel **04**

(d) Describe briefly salient features of total station. **03**

OR

Q - 3 (a) **Define the following** **02**

a) Point of tangency
b) Vertical curve
c) Trough compass
d) Johnson's table

(b) What are the methods of plane tabling? Explain the method of traversing. **04**

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(c) It is desired to set out a rectangular building 40m x 36m with 15 cm thick walls. The width of foundation is 1m. The building is divided into three equal parts. The internal walls are 10 cm thick and foundation width is 75cm. Draw a detail plan showing all pegs required for setting out building. **04**

(d) Differentiate varnier theodolite and optical theodolite. **04**

SECTION - II

Q - 4 (a) Describe with the help of sketches, various characteristics of the contours. **04**

(b) Explain the vertical control for a hydrographic survey. **03**

Q - 5 (a) Derive an expressions to compute the area from offsets to a base line by **07**

a) Trapezoidal rule and

b) Simpson's one third rule

(b) What is longitudinal sectioning? Write field procedure of longitudinal sectioning. **03**

(c) The following offsets are taken from a survey line to a curve boundary line **04**

Distance(M)	0	5	10	15	20	30	40	60	80
Offset (M)	2.4	3.7	4.5	5.3	6	4.8	5.7	3.8	2.1

Find out area between survey line, the curved boundary line and the first and last offset by trapezoidal rule.

OR

Q – 5 (a) The ground level along the centre line of a road is given below **05**

Chainage(M)	0	50	100	150	200	250	300
Ground Level (M)	218.0	216.50	216.00	216.75	217.30	217.95	215.65

It is proposed that formation level of reduced level 215.00 should be kept constant of starting from the chainage "zero", the formation width of the road is 7.0m and side slope 1:1, The ground is level transverse to the centre line.

(b) Explain triangle method of computing area of traverse. **02**

(c) How permanent adjustment of level is carried out? **03**

(d) Write a note on prismoidal correction and curvature correction. **04**

Q – 6 (a) What is sounding? What are the methods of locating sounding? Explain location of soundings from boat. **04**

(b) ABCD is a closed traverse in which the length and bearing have not been observed the rest of the field record is as follows: **05**

Line	Length (M)	Bearing
AB	235.10	338°20'
BC	317.40	82°22'
CD	215.00	167°00'
DA	?	?

Calculate the length and bearing of line DA.

(c) Define the following terms **05**

a) closing error

b) latitude of a line

c) relative closing error

d) consecutive co-ordiante

e) independent co-ordinate

OR

Q – 6 (a) What is hydrographic survey? What are the main purposes for which hydrographic surveying is carried out? **04**

(b) Define the following terms of vernier transit theodolite **05**

a) upper plate
b) transiting or plunging
c) face right
d) verticle circle
e) line of collimation

(c) Prepared a Gale's traverse to adjust the closing error of the closed traverse ABCDA for the following data: **05**

Line	Length (M)	Bearing
AB	110.00	110°00'
BC	80.00	170°00'
CD	95.00	250°00'
DA	160.00	350°00'

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